

**Epistemic Lensing:
A Markov-Blanket Account of
Mediated Belief Distortion**

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Abstract

This paper proposes *epistemic lensing* as a formal concept for describing how mediated information channels deform belief formation. Using a Markov-blanket framing, we distinguish direct environmental evidence from processed informational inputs and define distortion as divergence between the posterior supported by a higher-fidelity channel and the posterior induced by a mediated one. We introduce five distortion operators (attenuation, selection, warping, amplification, recursion) and four families of metrics: information loss, posterior divergence, inferential curvature, and hysteresis. A toy simulation shows how channel structure produces systematic overreaction, underreaction, salience warping, and persistent miscalibration in both individual agents and populations. We extend the framework to populations and introduce *posterior shear*, *focal capture*, *shadow zones*, and *narrative gravity wells* as collective-level phenomena. The framework reads as a bridge between active inference, information theory, media ecology, and political cognition.

1. Introduction

The standard vocabulary for describing epistemic failure in public life is impoverished. We speak of citizens as “low information” or “high information,” as though the relevant variable were volume. We speak of “misinformation” and “disinformation,” as though the problem were discrete false propositions injected into an otherwise clean signal. We speak of “media bias,” as though distortion were a scalar that could be placed on a left-right axis and corrected by balance.

None of these framings capture the actual structure of the problem. People do not simply have more or less information. They inhabit *different inferential geometries*. Two agents exposed to the same underlying reality, through differently structured channels, will disagree on more than facts. They will disagree about what counts as evidence, how strongly evidence bears on conclusions, and which aspects of reality are salient at all. The real question is not volume of information but *shape of transformation*.

Ignorance is one shape of the pathology. Another is guided misperception under conditions of structured mediation.

This paper proposes a framework for making the second claim precise. The central concept is *epistemic lensing*: the systematic deformation of belief-updating induced by the channel between world and agent. The optical metaphor is deliberate. A lens does not block light; it bends it. A mediating channel does not necessarily reduce information; it *reshapes* the inferential path from evidence to posterior. Some reshaping is corrective: journalism, expertise, and synthesis often *increase* fidelity relative to raw sensory exposure. Distortion begins when the channel bends posterior beliefs away from what a higher-fidelity channel would support.

The framework rests on three components. The first is a *three-layer architecture* (world, channel, posterior) formalized through Markov blankets. The second is a *taxonomy of distortion operators* that describe the elementary ways a channel can reshape inference. The third is a *family of metrics* that make distortion measurable. We demonstrate the framework with a toy Bayesian simulation and extend it to populations, where channel structure induces collective phenomena (polarization,

epistemic fragmentation, persistent miscalibration) that no account of individual cognitive failure alone can explain.

2. Theoretical Framework

2.1 The Three-Layer Architecture

Agents do not encounter the world directly. They encounter it through channels. Mediation is the basic architecture of bounded cognition rather than a pathological condition. An organism's sensory apparatus is itself a channel: one that compresses, selects, and transforms environmental signals into a format the organism can process. The question is never whether mediation occurs, but what *kind* of mediation, and with what consequences for inference.

We formalize this with a three-layer picture:

- **World** W_t : the state of affairs at time t , including events, causal structure, distributions over observables.
- **Evidence stream** X_t : a relatively high-fidelity signal from W_t . Not “raw reality” (which may be inaccessible), but the best evidence reasonably available.
- **Mediated stream** M_t : the signal that actually reaches the agent after processing by a mediating channel $\mathcal{C}$.

The agent forms a posterior belief $q(W_t \mid \cdot)$ based on whichever stream it receives. We distinguish:

- $q^*(W_t \mid X_t)$: the *benchmark posterior*, what a well-calibrated agent would believe given high-fidelity evidence.
- $q_i(W_t \mid M_t)$: agent i 's *mediated posterior*, what they actually come to believe given the channel output.

2.2 Markov-Blanket Framing

The architecture has a correspondence in the Markov-blanket formalism (Pearl, 1988; Friston, 2013). In the language of active inference, an agent is separated from its environment by a Markov blanket: sensory states and active states that mediate all interaction between internal states and external states. The agent's internal model of the world is updated only through blanket-facing inputs.

A mediating channel $\mathcal{C}$ interposes between the environment and the blanket. It transforms the evidence stream before it reaches the agent's sensory surface:

$$M_t = \mathcal{C}(X_t, \theta_{\mathcal{C}})$$

where $\theta_{\mathcal{C}}$ parameterizes the channel's structure: noise level, selection function, amplification profile, and so on. The agent does not observe $\mathcal{C}$ directly; it observes only M_t and updates accordingly. From the agent's perspective, M_t is the evidence.

2.3 Definition

Definition 1 (Epistemic Lensing). *Epistemic lensing occurs when the mapping $W_t \rightarrow M_t \rightarrow q_i$ systematically deforms belief relative to the benchmark mapping $W_t \rightarrow X_t \rightarrow q^*$. Formally, a channel $\mathcal{C}$ induces epistemic lensing with respect to benchmark q^* if, for a non-negligible set of world states and time periods:*

$$D(q_i(W_t | M_t), q^*(W_t | X_t)) > \epsilon$$

where D is a divergence measure on probability distributions and $\epsilon > 0$ is a calibration tolerance.

The benchmark q^* is the posterior that a well-calibrated agent would form given a *higher-fidelity* channel, with no claim to omniscience or to direct access to W_t . The framework measures *relative* fidelity, leaving aside metaphysical access to the thing-in-itself.

Mediation is not always distortion. A channel that compresses redundant signals, filters noise, or synthesizes scattered evidence may *improve* calibration relative to raw exposure. The concept of epistemic lensing targets the *deformative* component: the residual divergence after accounting for any corrective value the channel provides.

The distortion must be *systematic*. Random errors wash out under repeated observation. Epistemic lensing describes *structured* deformation: consistent directional bias, systematic salience warping, or persistent occlusion of specific domains.

3. Distortion Operators

A channel transmits reality and, in transmitting, edits the geodesics by which reality reaches the posterior. We identify five elementary operations by which a channel $\mathcal{C}$ can reshape inference. Any real mediating system can be decomposed as a composition of these operators.

3.1 Attenuation $\mathcal{A}$

Attenuation removes signal. The channel transmits less information than it receives:

$$I(W_t; M_t) < I(W_t; X_t)$$

where $I(\cdot; \cdot)$ is mutual information. Attenuation does not bias inference in a particular direction; it increases uncertainty. The agent's posterior becomes wider, less confident, less precise. In the limit, total attenuation produces maximum entropy: the agent knows nothing.

Examples: news deserts (geographic attenuation of local information), paywalls (economic attenuation), language barriers, algorithmic deprioritization of low-engagement content.

3.2 Selection $\mathcal{S}$

Selection passes some signals and blocks others. Where attenuation reduces signal uniformly, selection operates on *content*:

$$M_t = \mathcal{S}(X_t) = X_t \cdot \mathbf{s}$$

where $\mathbf{s} \in \{0, 1\}^n$ is a selection mask over n dimensions of the evidence space. The agent receives a *partial* world-model: accurate in the domains the channel transmits, absent in the domains it blocks.

Selection is particularly dangerous because it is invisible to the agent. The agent cannot distinguish between “nothing is happening in domain d ” and “the channel does not transmit domain d .” This asymmetry between absence-of-evidence and evidence-of-absence is the epistemic core of selection bias.

Examples: editorial selection of stories, algorithmic content curation, geographic isolation from certain populations or events, institutional specialization.

3.3 Warping $\mathcal{W}$

Warping reframes content. The channel goes beyond selecting which signals to pass; it transforms their *meaning* relative to the agent’s inferential framework:

$$M_t = \mathcal{W}(X_t) = f_{\mathcal{W}}(X_t)$$

where $f_{\mathcal{W}}$ is a nonlinear transformation that preserves some structural features of X_t while distorting others. Warping produces directional bias: the posterior is *shifted*, pulled toward specific regions of belief-space rather than simply broadened or narrowed.

Examples: partisan framing, euphemism and dysphemism, context-stripping, misleading juxtaposition, emotive imagery selection.

3.4 Amplification $\mathcal{G}$

Amplification overweights certain cues relative to their evidential strength:

$$M_t^{(j)} = g_j \cdot X_t^{(j)}, \quad g_j > 1 \text{ for some } j$$

where g_j is a gain factor on dimension j of the evidence. Amplification produces *salience inflation*: the agent treats certain signals as more informative than they are. If the amplified cues happen to be diagnostic, performance improves. If they are not, calibration degrades because posterior mass concentrates on a narrow subset of the evidence.

Examples: sensationalist coverage of rare events, trending algorithms that promote high-engagement content regardless of base rate, availability heuristic exploitation.

3.5 Recursion $\mathcal{R}$

Recursion occurs when the channel's output feeds back into itself:

$$M_{t+1} = \mathcal{C}(X_{t+1}, M_t)$$

The channel at time $t + 1$ depends on new evidence and on its own prior output. This creates *path dependence*: the agent's beliefs at t shape the channel's behavior at $t + 1$, which shapes beliefs at $t + 2$, and so on. Recursion is the operator responsible for *hysteresis*, the persistence of distorted beliefs after corrective evidence arrives, and for *lock-in*, where initial channel biases compound over time.

Examples: recommendation algorithms that personalize content based on prior engagement, echo chambers, confirmation-seeking information behavior, viral amplification of early framings.

3.6 Composition

Real mediating systems compose these operators. A social media feed might simultaneously *select* (algorithmic curation), *amplify* (engagement-weighted ranking), *warp* (framing by content creators), and *recurse* (personalization based on prior behavior). The distortion of the composite channel is the *interaction* of individual distortions rather than their sum, and the interaction can be nonlinear. A mildly selective channel with strong recursion may produce more distortion than a heavily warped channel without feedback.

4. Metrics of the Bend

Epistemic lensing becomes a scientific concept once distortion is measurable. We introduce four families of metrics, each capturing a different aspect of how the channel deforms inference.

4.1 Information Loss

How much world-relevant information survives mediation?

$$\mathcal{L} = 1 - \frac{I(W_t; M_t)}{I(W_t; X_t)}$$

$\mathcal{L} = 0$ means the channel preserves all world-relevant information, even if it warps or amplifies. $\mathcal{L} = 1$ means total attenuation. Information loss measures the *quantity* of evidential signal destroyed without regard to its direction.

Low information loss does not imply low distortion. A channel can preserve mutual information while systematically warping the *structure* of the signal. Information loss is a necessary but not sufficient indicator.

4.2 Posterior Divergence

How far does the mediated posterior bend from the benchmark?

$$\mathcal{D} = D_{JS}(q_i(W_t | M_t), q^*(W_t | X_t))$$

where D_{JS} is the Jensen-Shannon divergence: symmetric, bounded, well-defined even when the distributions do not share support. Posterior divergence is the most direct measure of epistemic lensing; it compares what the agent believes to what they would believe under a better channel.

Posterior divergence decomposes into components: *location shift* (the posterior mean moves), *dispersion change* (the posterior width changes), and *shape distortion* (the posterior geometry changes in ways not captured by location and scale). The decomposition distinguishes, for instance, an agent who is confidently wrong (low dispersion, high location shift) from one who is merely confused (high dispersion, low location shift).

4.3 Inferential Curvature

The third metric is the most novel one in the family. It compares the *sensitivity* of belief-update to evidence under the two channels rather than the *level* of belief.

Let e denote an increment of evidence. Define the *update function*:

$$u(e) = \frac{\partial q(W_t)}{\partial e}$$

Inferential curvature is the difference in update sensitivity between mediated and benchmark channels:

$$\kappa = u_{\text{mediated}}(e) - u_{\text{benchmark}}(e)$$

Three regimes:

- $\kappa > 0$: **magnification**. The agent updates too much from evidence of a given strength, treating weak signals as if they were strong.
- $\kappa < 0$: **occlusion**. The agent updates too little, with strong signals failing to move belief.
- $\text{sign}(\kappa) \neq \text{sign}(u_{\text{benchmark}})$: **deflection**. The agent updates in the wrong direction, with evidence that should increase q instead decreasing it, or the reverse.

Inferential curvature catches something posterior divergence misses. Two agents can hold the *same* posterior at time t with *different* curvatures, meaning they will diverge in how they respond to future evidence. Curvature is a second-order property: it describes the geometry of the inferential surface rather than the current position on it.

4.4 Hysteresis

How much distortion persists after corrective evidence arrives?

Let E_{correct} be a corrective evidence event that, under the benchmark channel, would bring the posterior to within ϵ of q^* . Hysteresis is the residual divergence after correction:

$$\mathcal{H} = D_{JS}(q_i(W_t \mid M_{1:t}, E_{\text{correct}}), q^*(W_t \mid X_{1:t}, E_{\text{correct}}))$$

$\mathcal{H} = 0$ means perfect correction: the agent snaps back to the benchmark. $\mathcal{H} > 0$ means *epistemic residue*. The history of distortion leaves a trace that survives contact with the truth.

Hysteresis is the metric most directly relevant to democratic epistemology. A society can tolerate temporary distortion if correction is effective. When hysteresis is high, even perfect corrective information fails to undo accumulated damage: the bend has set into the agent's posterior and structures every later update.

5. Toy Model

5.1 Setup

We construct a minimal simulation to demonstrate that the framework produces quantitatively distinct patterns of distortion rather than only metaphorical descriptions. The model is deliberately simple: one world state, one agent, one channel, observed over time.

World. A continuous latent state $W_t \in \mathbb{R}$ evolving as a random walk:

$$W_{t+1} = W_t + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \sigma_W^2)$$

Evidence stream. The high-fidelity evidence is the world state plus observation noise:

$$X_t = W_t + \xi_t, \quad \xi_t \sim \mathcal{N}(0, \sigma_X^2)$$

Mediated stream. The channel applies a parameterized transformation:

$$M_t = (1 - \gamma)(\mathcal{S}(X_t) + \beta \cdot b) + \gamma \cdot M_{t-1} + \nu_t$$

where:

- $\mathcal{S}$ is a selection function that drops the signal with probability p_{omit} (on a dropped step the agent receives channel noise carrying no information about W_t and runs no update),
- β controls warping (injection of directional bias b),
- $\gamma \in [0, 1)$ controls recursion (dependence on prior channel output),
- $\nu_t \sim \mathcal{N}(0, \sigma_\nu^2)$ is channel noise.

Attenuation is modeled as a reduction in signal-to-noise: a parameter $\alpha \in (0, 1]$ adds $(1/\alpha - 1) \sigma_X^2$ of variance, so the channel still tracks X_t with no directional bias but transmits less information. Modeling attenuation as a multiplicative shrinkage of the mean was avoided because that injects the directional bias the framework assigns to warping. Amplification is modeled by scaling the surprising part of the signal, the innovation $X_t - X_{t-1}$, by a gain factor $g > 1$ whenever that jump exceeds a threshold τ , so the operator inflates rare events rather than rescaling the level.

Agent. A Bayesian updater (a one-dimensional Kalman filter on the random-walk prior) maintaining a Gaussian posterior $q_t = \mathcal{N}(\mu_t, \sigma_t^2)$, with Kalman gain $k_t = \sigma_t^2 / (\sigma_t^2 + \sigma_{\text{obs}}^2)$:

$$\mu_t = k_t \cdot M_t + (1 - k_t) \cdot \mu_{t-1}$$

Crucially, the agent does not observe the channel $\mathcal{C}$ (Section 2.2): it uses a fixed assumed observation noise σ_{obs} and treats M_t as direct evidence. It cannot widen its posterior to account for a degraded channel it cannot see. The same agent run on the high-fidelity stream X_t supplies the benchmark posterior q^* ; run on M_t it supplies the mediated posterior q_i . All distortion is measured as q_i relative to q^* .

5.2 Scenarios

We run the model under four channel configurations, each isolating a different distortion operator, with all other parameters at baseline. Each metric is averaged over 400 seeded repetitions; the run is reproducible from a single command and writes `simulation/output/results.json`.

Scenario A: Pure attenuation. $\alpha = 0.3$, all other distortion parameters zero. The agent receives a low signal-to-noise signal. *Hypothesis:* tracking degraded, with no systematic directional pull.

Scenario B: Selection + warping. $p_{\text{omit}} = 0.5$, $\beta = 0.4$ (positive bias). The agent misses half the evidence and receives a directionally biased remainder. *Hypothesis:* posterior shifted toward b , with intermittent tracking when transmitted signals arrive.

Scenario C: Amplification. $g = 3$ on innovations with $|X_t - X_{t-1}| > \tau$ (rare jumps amplified). *Hypothesis:* salience inflation, the agent overreacts to surprising signals. Posterior oscillates more than q^* .

Scenario D: Recursion. $\gamma = 0.7$ (strong autoregressive feedback). *Hypothesis:* the channel mixes old and new signal, so the posterior tracks W_t with a lag.

5.3 Results

All values below are computed by the simulation and read directly from `results.json`. Divergences are Jensen-Shannon, in bits, bounded in $[0, 1]$.

Information loss ($\mathcal{L}$): Highest under selection + warping (Scenario B, $\mathcal{L} = 0.88$), where dropping half the steps destroys the most mutual information. Amplification (C, $\mathcal{L} = 0.68$) is also high,

because scaling rare innovations decorrelates M_t from W_t . Pure attenuation is moderate (A, $\mathcal{L} = 0.37$). Recursion is lowest (D, $\mathcal{L} = 0.16$): the channel preserves the signal and only delays it.

Posterior divergence ($\mathcal{D}$): Highest under amplification (C, $\mathcal{D} = 0.36$). The other three cluster tightly (A = 0.22, B = 0.24, D = 0.24), so the scalar alone understates how differently the channels fail. The decomposition into location shift and dispersion change (Section 4.2) separates them. Every channel’s divergence is dominated by location shift (A = 0.86, B = 1.07, C = 1.39, D = 0.92); only selection + warping also widens the posterior (dispersion change = 0.44), because dropped steps leave the agent genuinely less certain. The other three show a dispersion change of 0.00: the agent stays just as confident while being wrong, since it cannot see the channel and so never widens to compensate. This is the formal content of *confidently wrong*.

Inferential curvature (κ): Near zero under attenuation (A, $\kappa = -0.00$): the agent tracks at the right sensitivity, only noisier. Strongly positive under amplification (C, $\kappa = +1.06$): magnification, the agent updates far more than the evidence warrants. Negative under selection + warping (B, $\kappa = -0.27$) and recursion (D, $\kappa = -0.43$): occlusion, the agent updates too little, because dropped steps and stale channel memory both blunt the response to new evidence. Curvature is the metric that most cleanly separates the operators, even where posterior divergence does not.

Hysteresis ($\mathcal{H}$): At $t = 150$ the world sharply reverses direction; the corrective evidence then arrives through each channel rather than by oracle, and we measure the residual divergence over the following 10 steps. Amplification shows the largest residue (C, $\mathcal{H} = 0.35$) because the reversal is itself a large innovation that the channel over-amplifies. Selection + warping (B, $\mathcal{H} = 0.25$) and recursion (D, $\mathcal{H} = 0.24$) follow, attenuation last (A, $\mathcal{H} = 0.22$). One honest finding: in this minimal single-agent model the post-reversal residual tracks each channel’s ongoing steady-state distortion rather than isolating a memory-specific residue. Recursion lags, but a single reversal does not make it uniquely uncorrectable; lock-in compounds under sustained recursion across a population, which a one-shot correction leaves untouched (Section 6.5).

5.4 Key Finding

The model makes the framework’s central distinction quantitative: ignorance and distortion are different failures with different signatures. Attenuation is ignorance: high information loss (0.37), curvature near zero, divergence carried by random location error with no confident bias. The agent is wrong only by being noisy. Amplification and selection-plus-warping are distortion: amplification combines the highest divergence (0.36) with strong positive curvature (+1.06), selection the highest information loss (0.88) with negative curvature (−0.27). The decomposition is the sharpest evidence: three of the four channels distort the posterior’s location while leaving its width unchanged, so the agent is confident and wrong at once. The blanket assumption that the agent cannot see its channel is what makes confident error, rather than honest uncertainty, the default response to a degraded signal.

6. Population Extension

6.1 From Individual to Field

The framework extends naturally from a single agent to a population. Suppose N agents, each receiving W_t through a potentially different channel $\mathcal{C}_i$ with parameters $\theta_{\mathcal{C}_i}$. The population's belief state is the *posterior field*: the distribution of posteriors $\{q_1, q_2, \dots, q_N\}$ across agents.

Public distortion shows up at the population level as a warped inferential field across many agents, beyond the false belief visible at the individual level.

6.2 Posterior Shear

When subpopulations are exposed to differently structured channels, their posteriors diverge systematically. We define *posterior shear* as:

$$\Sigma = D_{JS}(\bar{q}_A, \bar{q}_B)$$

where $\bar{q}_A$ and $\bar{q}_B$ are the average posteriors of groups A and B . High shear means the groups inhabit *different epistemic worlds*. The cause is structural rather than dispositional: the channels deliver differently shaped evidence, regardless of whether the groups share values or priors.

Posterior shear differs from polarization in a standard sense. Two groups can be polarized because they weight evidence differently (a cognitive phenomenon). Posterior shear says the groups are polarized *because the evidence that reaches them is differently structured* (a channel phenomenon). The distinction matters for intervention: cognitive depolarization strategies (perspective-taking, deliberation) will fail if the underlying channel structure remains divergent.

6.3 Focal Capture

When a channel amplifies a narrow set of signals, collective attention collapses around those signals. *Focal capture* occurs when:

$$H(\text{Attention}(t)) < H_{\text{benchmark}}$$

where H is the entropy of the attention distribution over topics. The population attends to fewer topics than the world's complexity warrants. The captured topics may be important. Focal capture means that other important topics fall below the threshold of collective visibility.

6.4 Shadow Zones

The dual of focal capture is *shadow zones*: regions of the world that become effectively invisible because the channel rarely transmits them. A shadow zone is a topic that the channel *could* transmit

and *does not*, either because it fails selection criteria (judged unnewsworthy or low-engagement) or because it is actively suppressed.

Shadow zones are epistemically dangerous because they are invisible. An agent in a shadow zone does not know they are missing information; the channel provides no signal that a signal is absent.

6.5 Narrative Gravity Wells

The most complex population phenomenon. A *narrative gravity well* is a dominant explanatory frame that pulls heterogeneous evidence into a single attractor basin. Once established, the narrative re-interprets incoming evidence to be consistent with itself:

- Confirming evidence is assimilated directly.
- Disconfirming evidence is reframed (“that’s exactly what they would say”).
- Ambiguous evidence is recruited (“this proves it’s even deeper than we thought”).

Narrative gravity wells combine warping (evidence is reframed), amplification (confirming cues are overweighted), and recursion (the narrative feeds back into itself). They are stable because they are self-reinforcing, and they produce high hysteresis, since corrective evidence is absorbed rather than processed.

7. Empirical Program

The framework suggests a structured empirical research program at three levels.

7.1 Individual Level

At the individual level, epistemic lensing predicts measurable differences in belief calibration, update sensitivity, and correction response as a function of channel structure. The relevant variables are:

- **Calibration error:** the gap between an agent’s confidence and their accuracy, measured across domains.
- **Update sensitivity:** how much an agent’s posterior shifts in response to standardized evidence, compared to a Bayesian benchmark.
- **Correction persistence:** whether belief revision from corrective information endures over time, or whether prior distortion reasserts itself.

Experimental designs would expose participants to identical world-state information through differently structured channels and measure downstream belief, confidence, and update behavior.

7.2 Channel Level

At the channel level, the framework predicts that measurable properties of mediating systems (concentration, diversity, repetition, ranking logic, cue distribution) should predict population-level distortion metrics. The relevant variables are:

- **Source concentration:** the Herfindahl index of information sources within a population.
- **Signal diversity:** the entropy of topic coverage relative to a benchmark of world-state dimensionality.
- **Recursion depth:** the degree to which a channel's output at t depends on its output at $t - 1$ (measurable in recommendation algorithms via autoregressive analysis of content feeds).
- **Correction latency:** the time delay between a world-state change and the channel's transmission of the corrective signal.

7.3 Place Level

At the geographic level, epistemic lensing predicts that local media ecology should predict local belief distortion. The relevant variables are:

- **Broadband access and platform penetration:** which channels are available.
- **Network homophily:** how structurally similar an agent's information network is.
- **Institutional trust:** baseline trust in mediating institutions (journalism, government, science) conditions how agents weight channel outputs.
- **Channel plurality:** the number of structurally independent information sources available in a locale.

The framework treats some mediation as corrective. Journalism, expertise, and synthesis often increase fidelity. The object of study is the subset of processing that worsens calibration.

8. What a comparative benchmark can and cannot fix

The benchmark is approximate. We define distortion relative to a higher-fidelity channel rather than to unmediated reality. In practice, determining which channel is "higher-fidelity" requires judgment and itself involves epistemic assumptions. The framework operates inside the hermeneutic circle, using relative fidelity as a tractable proxy.

Direct reality is often inaccessible. For many questions of public concern (the state of the economy, the trajectory of a pandemic, the effects of a policy), no agent has direct access to W_t . All channels are mediated to some degree. The framework is comparative: it asks which mediation structures produce better-calibrated posteriors, rather than whether any structure achieves perfect calibration.

Humans necessarily rely on mediation. A framework that implied citizens should bypass all channels and encounter the world directly would be both impractical and incoherent. Most knowledge is necessarily second-hand. The question is how to recognize when a channel's structure is deforming inference rather than supporting it.

The framework does not specify normative content. Epistemic lensing describes *how* beliefs are deformed and stays silent on *which beliefs are correct*. It is a structural analysis. It can be applied symmetrically: any channel, regardless of its ideological orientation, can be assessed for information loss, posterior divergence, inferential curvature, and hysteresis. Whether channels are in fact

symmetrically distortive is then an empirical question the metrics can settle rather than assume; work on asymmetric misinformation ecosystems (Benkler, Faris & Roberts, 2018) argues they are not, and the framework offers a way to quantify the asymmetry instead of stipulating it.

The toy model is a logic demonstration. The numbers in Section 5 are computed and reproducible, but the Bayesian agent serves as a normative benchmark rather than a model of actual human cognition. Real agents exhibit motivated reasoning, identity-protective cognition, and bounded rationality that interact with channel structure in complex ways. The model demonstrates the *logic* of distortion; empirical validation requires the program outlined in Section 7.

9. Ignorance and distortion are different problems

The distance between citizens and reality is the familiar diagnosis. The harder problem is that the route from reality to belief may have been curved in systematic ways.

This paper has proposed *epistemic lensing* as a framework for describing, measuring, and analyzing that curvature. The framework distinguishes five elementary distortion operators (attenuation, selection, warping, amplification, recursion) and four metric families (information loss, posterior divergence, inferential curvature, hysteresis). It extends from individual agents to populations, where channel structure induces collective phenomena: posterior shear, focal capture, shadow zones, and narrative gravity wells.

The framework's central payoff is a distinction between *ignorance* and *distortion*. Ignorance is a deficit of signal. Distortion is a reshaping of the inferential path. The two have different causes, dynamics, metrics, and remedies. A society suffering from ignorance needs more information. A society suffering from distortion needs *differently structured* channels, and the institutional capacity to recognize when its channels are bending inference rather than supporting it.

Modern publics live inside information optics. Some channels clarify, compress, and correct. Others bend, magnify, occlude, and lock belief into persistent trajectories that later evidence struggles to undo. The political question is not only who knows what, but through what curvature reality is allowed to arrive.

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